

Forensic Odontological Parameters as Biometric Tool: A Review.

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Human identification by virtue of dental identification is on the increase in the past few years and has served as a major adjunct in the identification process. The science associated with the dealing of evidences from dental, oral, and paraoral structures-Forensic Odontology, is a boon in itself. It has been seen that the role of forensic odontology has increased as most of the times teeth and other orally associated structures remain the only means of identification in the worst of scenarios where the conventional biometrics such as fingerprint, iris scan, palm print, etc., may be lost and are therefore not reliable enough to resolve the purpose of identification. There are various important aspects in forensic odontology which are associated with the study of tooth prints, i.e., amelogyphics, radiographs, the study of palatal rugae patterns, i.e., rugoscopy, cheiloscopy, i.e., the study of lip patterns, tongue prints which can serve as an important biometric tool to aid in the human identification process. Dental biometrics helps to determine several parameters that serve in the interest of forensics such as type of specimen, population affinity, sex, age, stature, race, and various other individualization factors. So, comparative dental identifications with the help of a biometric recognition system will play a major role in identifying deceased individuals in disasters, crimes, or any other mass tragedies. How to cite this article: Kaul B, Vaid V, Gupta S, et al. Forensic Odontological Parameters as Biometric Tool: A Review. Int J Clin Pediatr Dent 2021;14(3):416-419.

Are Tooth Prints a Hard Tissue Equivalence of Finger Print in Mass Disaster: A Rationalized Review.

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Personal identification methods may not be efficient when bodies are decomposed, burned, in cases of mass disasters when soft tissue cannot provide reliable information or has been lost. Various methods currently employed in forensic odontology for personal identification include comparing with antemortem dental charts, rugoscopy, denture labeling, DNA analysis from dental pulp, bite mark analysis, etc., Recently, there is growing interest in the study of enamel rod end patterns. These enamel rod end patterns are termed as "Tooth prints" and the study of these prints is known as "Amelogyphics" (amelo: Enamel, Glyphics: Carvings). This review encompasses about the basis of using enamel rod end patterns, methods of obtaining the patterns and further suggests these tooth prints as an analogy to finger print in personal identification in mass disasters.

Experimental studies of forensic odontology to aid in the identification process.

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The importance of dental identification is on the increase year after year. With the passage of time, the role of forensic odontology has increased as very often teeth and dental restorations are the only means of identification. Forensic odontology has played a key role in identification of persons in mass disasters (aviation, earthquakes, Tsunamis), in crime investigations, in ethnic studies, and in identification of decomposed and disfigured bodies like that of drowned persons, fire victims, and victims of motor vehicle accidents. The various methods employed in forensic odontology include tooth prints, radiographs,

photographic study, rugoscopy, cheiloscopy and molecular methods. Investigative methods applied in forensic odontology are reasonably reliable, yet the shortcomings must be accounted for to make it a more meaningful and relevant procedure. This paper gives an overview of the various experimental studies to aid in the identification processes, discussing their feasibilities and limitations in day-to-day practice.

Amelogyphics: A possible forensic tool for person identification following high temperature and acid exposure.

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Forensic odontology is a branch that is evolving over time and has opened newer avenues that may help in the identification of individuals. Tooth prints are the enamel rod end patterns on tooth surface and they are considered as a hard tissue analog to fingerprints. Teeth have the highest resistance to most environmental effects like fire, desiccation, and decomposition, and may be used as a forensic evidence. The aim of the study was to evaluate if the tooth prints could be used for an individual's identification and reproducibility and permanency of these tooth prints after exposing the teeth to acid and various degrees of temperature. 90 tooth prints from 20 freshly extracted maxillary premolar teeth were obtained. Cellophane tape technique was used to record enamel rod end patterns on tooth surface. Ten teeth (one from each patient) were immersed in 36.46% hydrochloric acid and the tooth prints were obtained at various intervals (5 min, 10 min, and 20 min). The other 10 teeth (one from each patient) were incinerated and impression was made at various intervals (80° C, 400° C, 600° C, and 750° C). Tooth prints obtained from different teeth (total of 90 tooth prints) were analyzed using Verifinger® standard SDK version 5.0 software. All the 20 original tooth prints were distinct from each other and no inter-individual or intra-individual similarity was found. The tooth prints from the same tooth after it was exposed to acid or heat were reproducible and showed high to very high similarity with the original tooth print of that particular tooth stored in the database. Tooth prints may be used as an effective aid in person identification even in adverse conditions such as burn and acid attack injuries.

Amelogyphics: A Forensic Tool for Human Identification.

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Crime rates in our contemporary technological age have been rising at an alarming pace. Newer methods are needed to determine who the perpetrator and victim are in these situations. By comparing a person's recorded information with their own unique characteristics, automated biometric systems may determine who they are. Biometric identifiers like fingerprint and iris scanning are among the most reliable ways. Due to the vulnerability of soft tissues, these procedures cannot be depended on in the event of widespread calamities such as fires or gas leaks. Therefore, improved individual identification necessitates a biometric system based on the hard tissue. Thus, amelogyphics was created to serve as a key tool in forensic odontology, particularly in the case of huge fatalities caused by natural catastrophes.

Dental print media and their value in forensic odontology.

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Teeth are one of the most important factors in the identification of unknown cadavers. One of the most efficient methods the police have of publicizing a case is to publish the victim's dental x-rays of treatment records in the dental print media which dentists will often recognize and thus be able to contribute to the identification. To evaluate the efficiency of this procedure, 177 cases published between 1975 and 1995 were reviewed. The results show that only 3% of the cases were identified from recognition in the dental print media. A survey of 500 dentists was carried out to gauge the efficacy of the presentation of identification cases in their dental print media. According to these results a revised method of presentation needs to be developed to improve the efficiency of this odontoforensic publicity medium. New technologies could also open new avenues for forensic identification such as digital x-raying or online data transfer. The Internet itself could be useful regarding international cooperation of forensic odontologists in the identification of unknown corpses.

The connecting link! Lip prints and fingerprints.

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Lip prints and fingerprints are considered to be unique to each individual. The study of fingerprints and lip prints is very popular in personal identification of the deceased and in criminal investigations. This study was done to find the predominant lip and fingerprint patterns in males and females in the North Indian population and also to find any correlation between lip print and fingerprint patterns within a gender. Two hundred students (100 males, 100 females) were included in the study. Lip prints were recorded for each individual using a dark-colored lipstick and the right thumb impression was recorded using an ink pad. The lip prints and fingerprints were analyzed using a magnifying glass. The Chi-square test was used for statistical analysis. The branched pattern in males and the vertical pattern in females were the predominant lip print patterns. The predominant fingerprint pattern in both males and females was found to be the loop pattern, followed by the whorl pattern and then the arch pattern. No statistically significant correlation was found between lip prints and fingerprints. However, the arch type of fingerprint was found to be associated with different lip print patterns in males and females. Lip prints and fingerprints can be used for personal identification in a forensic scenario. Further correlative studies between lip prints and fingerprints could be useful in forensic science for gender identification.

Effects of palivizumab prophylaxis on respiratory syncytial virus (RSV) infections in Montenegro.

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Respiratory syncytial virus (RSV) is one of the most common pathogens causing severe lower respiratory tract disease in infancy and childhood. In newborns, young infants, and in infants with co-morbidities, the risk of severe infection is increased. Current protection against severe RSV infection is immunoprophylaxis with the monoclonal antibody palivizumab. The study aimed to assess the effects of palivizumab prophylaxis in the Republic of Montenegro in comparison to the pre-prophylaxis period. The study was conducted in prospective/retrospective single center format in Montenegro in the Clinical Center of Podgorica, for the period 2009-2019. A total of 104 high-risk infants in the palivizumab prophylaxis program (2014-2019 RSV seasons) and 168 high-risk children without palivizumab

prophylaxis (2009-2013 RSV seasons) were enrolled. A total of 51 children (49.0%) received prophylaxis for prematurity, 33 (31.7%) for bronchopulmonary dysplasia (BPD), 13 (12.5%) for hemodynamically significant heart disease/defect (HSCHD), and 7 (6.8%) for "miscellaneous" indications. In the control group most children had prematurity (101, 60.1%), followed by BPD (59, 35.1%), HSCHD (3, 1.8%), and "miscellaneous" (5, 3.4%) conditions. Readmission to the pediatric intensive care units (PICU) due to RSV infection was significantly lower in prophylaxis group (0.0 vs 16.1%, $p < 0.001$). No lethal outcomes were observed in high-risk children with palivizumab prophylaxis compared to 2.4% in the control group. The introduction of RSV immunoprophylaxis as well as other new protective treatment strategies for high-risk newborns led to significant improvements in infant and childcare in Montenegro. This is the first report on palivizumab prophylaxis in Montenegro, demonstrating the effectiveness and safety of palivizumab use in clinical settings.

Cost-effectiveness of RSVpreF vaccine and nirsevimab for the prevention of respiratory syncytial virus disease in Canadian infants.

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Health Canada recently authorized the RSVpreF pregnancy vaccine and nirsevimab to protect infants against respiratory syncytial virus (RSV) disease. Assess the cost-effectiveness of RSVpreF and nirsevimab programs in preventing RSV disease in infants, compared to a palivizumab program. We used a static cohort model of a Canadian birth cohort during their first RSV season to estimate sequential incremental cost-effectiveness ratios (ICERs) in 2023 Canadian dollars per quality-adjusted life year (QALY) for nine strategies implemented over a one-year time period, from the health system and societal perspectives. Sensitivity and scenario analyses were conducted to explore the impact of uncertainties on the results. All-infants nirsevimab programs averted more RSV-related outcomes than year-round RSVpreF programs, with the most RSV cases averted in a seasonal nirsevimab program with catch-up. Assuming list prices for these immunizing agents, all-infants nirsevimab and year-round RSVpreF programs were never cost-effective, with ICERs far exceeding commonly used cost-effectiveness thresholds. Seasonal nirsevimab with catch-up for infants born outside the RSV season was a cost-effective program if prioritized for infants at moderate/high-risk (ICER $< \$28,000$ per QALY) or those living in settings with higher RSV burden and healthcare costs, such as remote communities where transport would be complex (ICER of $\$5700$ per QALY). Using a $\$50,000$ per QALY threshold, an all-infants nirsevimab program could be optimal if nirsevimab is priced at $\$110$ - 190 per dose and RSVpreF priced at $< \$60$ - 125 per dose. Prophylactic interventions can substantially reduce RSV disease in infants, and more focused nirsevimab programs are the most cost-effective option at current product prices.

Evolving Strategies for Respiratory Syncytial Virus (RSV): A Review Article of Preventive Agents and Vaccines for RSV.

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The objective was to describe the pharmacology, efficacy, safety, and recommendations for the use of newly approved preventive agents and vaccines for respiratory syncytial virus (RSV) and discuss their uptake during the 2023 to 2024 RSV season. A literature search of PubMed was performed (January 2020 to February 2024) with the search terms RSV vaccine, preventive antibody, and RSV prevention.

Utilization data were collected from TriNetX using the US Collaborative Network (May 2024) using the terms palivizumab, nirsevimab, and RSV prefusion F protein. Relevant English-language studies assessing the use of Food and Drug Administration (FDA)-approved preventive agents and vaccines for RSV in humans were considered. Population-level utilization data were extracted from TriNetX. Nirsevimab was observed to have noninferior efficacy and safety compared with palivizumab with less frequent administration. Nirsevimab is recommended to replace palivizumab for RSV prophylaxis in all eligible infants. Arexvy and Abrysvo are effective at reducing the risk of RSV infection in adults aged ≥ 60 years, and Arexvy is indicated in adults aged ≥ 50 years. These vaccines are equally recommended for use in the elderly adult population, but only Abrysvo is indicated and recommended for maternal administration. Most infants only require prophylaxis through either maternal RSV vaccination or nirsevimab administration.

Relevance to patient care and clinical practice: This review compares the indications for use, guideline recommendations, and clinical trial efficacy and safety data for palivizumab, nirsevimab, Abrysvo, and Arexvy to guide clinical decision-making. Novel RSV preventive agents, including Abrysvo, Arexvy, and nirsevimab, offer less burdensome dosing and administration compared with palivizumab, show promising efficacy and safety data, and expand the populations eligible for RSV prevention. Updated clinical guidance supports immediate adoption of these agents in practice, and population-level data suggest these agents were used during the 2023 to 2024 RSV season.
